

--	--	--	--	--	--	--	--

B.Tech. Degree I & II Semester Regular/Supplementary Examination April/September 2021

CS/IT 19-200-0203B ENGINEERING GRAPHICS

(2019 Scheme)

(Regular)

Time: 3 Hours

Maximum Marks: 60

PART A

(1 × 12 = 12)

- I. (a) Divide a line segment of 200 mm length into six equal parts. (4)
 (b) Construct a diagonal scale of RF 1/25 long enough to measure 4 meters. (8)
 Show a length of 2.86 m on it.

OR

- II. (a) Draw involute of a square 20 mm side. (4)
 (b) Construct a parabola in a parallelogram 100 mm and 60 mm sides with included angle of 70°. Locate the axis, double ordinate and focus. (8)

PART B

(4 × 15 = 60)

- III. (a) A line PQ 100 mm long is parallel to both HP and VP. The line is 40 mm below HP and 15 mm behind VP. Draw the front view and top view. (3)
 (b) Line AB which makes 30° to VP has the ends 10 mm and 40 mm above HP. The front view measures 50 mm. The vertical trace of the line is 15 mm below HP. Draw the views of the line and find out true length, inclination to HP and the horizontal trace of line AB. (12)

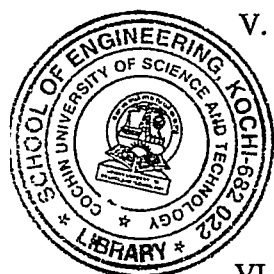
OR

- IV. (a) A hexagonal lamina 30 mm side lies on HP with one side making 20° to VP. Draw front view and top view keeping the lamina in first quadrant. (3)
 (b) A rectangular lamina 60 mm × 40 mm rests on HP on a shorter side which is making 35° to VP. Draw the front view and top view if the top view is a square. (12)

- V. (a) A square pyramid 40 mm side of base and 70 mm axis stands on VP on its base with an edge of base making 30° to HP. Draw the plan and elevation of the pyramid if the nearest corner of base is 20 mm above HP. (3)
 (b) An octahedron 50 mm side rests on HP on a triangular face with one side of this triangular face making 20° to VP. Draw the two views of the solid in this position. (12)

OR

- VI. (a) A cylinder 50 mm diameter and 70 mm axis stands on HP on its base. This is cut by a cutting plane so as to produce the biggest ellipse as the true shape of section. Draw the front and top views of the cylinder and show the position of cutting plane. What is the inclination of the cutting plane to HP? (3)
 (b) A pentagonal prism 25 mm side of base and 75 mm long lies on HP on a rectangular face with axis parallel to VP. It is cut by a vertical plane making 30° to VP and passing through a point on the axis 20 mm away from one of its ends. Draw sectional plan, sectional elevation and true shape of section. (12)



(P.T.O.)

BTS-I&II(R/S)-(04/09)-21-0808

- VII. (a) An equilateral triangle 100 mm side represents the total development of surface of a solid. Draw its plan and elevation in any convenient position. (3)
- (b) A hexagonal prism 30 mm side of base and 70 mm axis stands on HP on its base with an edge of base parallel to VP. A horizontal square hole of 32 mm side is drilled through the prism with its axis perpendicular to VP and all sides of the square equally inclined to HP in such a way that the axis of the hole bisects the axis of the prism. Develop the lateral surface of the prism with the hole drilled in it. (12)
- OR**
- VIII. A vertical square prism 50 mm side of base and 80 mm tall stands on HP on its base with one side of base making 30° to VP. Another square prism 36 mm side of base and 80 mm long penetrates the vertical square prism with its axis parallel to both planes and all lateral faces equally inclined to HP. Draw the three views of the arrangement showing the lines of intersection if the two axes intersect. (15)
- IX. (a) Draw isometric view of a pentagonal lamina 30 mm side kept parallel to VP. (3)
- (b) A hemisphere 70 mm diameter is kept on a circular slab 70 mm diameter and 25 mm thick. The hemisphere rests with its flat circular face on the slab so that their rims coincide. Draw isometric projection of the arrangement. (12)
- OR**
- X. (a) A circular lamina 50 mm diameter lies on the ground behind the picture plane touching picture plane. The station point is 70 mm in front of PP, 60 mm above ground plane and the central plane passes through 40 mm to the right of the centre of the circular lamina. (5)
- (b) A square prism 40 mm side and 70 mm long lies on the ground on a rectangular face with axis making 30° to PP and a vertical edge touching PP. The station point is 60 mm in front of PP, 55 mm above GP and the central plane passes through the centre of the prism axis. Draw the perspective view. (10)
